

Unsupervised Learning Assignment

Chapter 12 - Exercises 5 and 10

Group 2

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Course: Data Mining / Unsupervised Learning

1 Question 5: Effect of Scaling on K-Means Clustering

The exercise considers eight online shoppers and two variables: the number of socks purchased and the number of computers purchased. K-means with $K = 2$ is considered under three different scalings.

Table 1: Expected K-means behavior under the three scalings.

Scaling	Variable that dominates distance	Expected $K = 2$ result
Raw counts	purchase Socks	Clusters are driven mainly by differences in the number of socks. The shopper with the unusually high sock count is likely to be separated from the rest.
Standardized variables	Socks and computers contribute comparably	The computer-purchase pattern becomes important. A natural split separates shoppers who purchased a computer from those who did not.
Dollar spending	Computers	Because computers are far more expensive than socks, computer spending dominates Euclidean distance. The split is again mainly computer buyers versus non-buyers.

Explanation. K-means minimizes within-cluster squared Euclidean distances. Therefore, a variable with a larger numerical scale can dominate the clustering result. Standardizing the variables prevents one variable from dominating solely because of its scale. If the variables are instead expressed in dollars, computer purchases receive far greater weight because their monetary value is much larger.

2 Question 10: Simulated Data, PCA, and K-Means

(a) Generate three classes with 60 observations and 50 variables

A reproducible data set was generated with 20 observations in each of three classes. Every observation has 50 variables. To make the classes distinct, Class 1 was shifted in variables 1–10, Class 2 in variables 11–20, and Class 3 in variables 21–30. The shift was 4 units, while random noise followed a standard normal distribution.

```
rng = np.random.default_rng(42)
n_per_class = 20
p = 50
X_parts = []
true_labels = []

for c in range(3):
    mu = np.zeros(p)
    mu[c*10:(c+1)*10] = 4.0
    X_parts.append(rng.normal(loc=mu, scale=1.0,
```

```

        size=(n_per_class, p)))
    true_labels.extend([c + 1] * n_per_class)

X = np.vstack(X_parts)
y = np.array(true_labels)

```

The resulting matrix has dimension 60×50 , with 20 observations in each true class.

(b) PCA and the first two principal components

PCA was applied to the 60 observations. The first principal component explained 36.14% of the variance and the second explained 33.73%. Together, the first two components explained approximately 69.88% of the total variance.

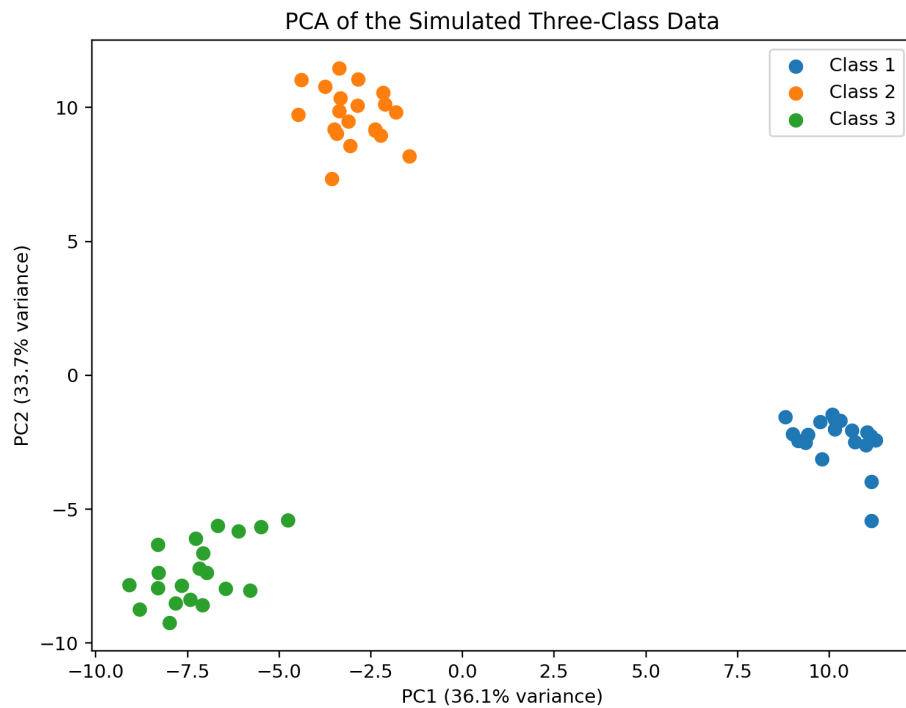


Figure 1: First two principal component scores, colored by the true class.

The three groups are clearly separated in the first two principal components, so the simulated data satisfy the requirement that the classes show visible separation before proceeding.

(c) K-Means with $K = 3$ on the raw data

K-means was fitted to all 50 original variables using $K = 3$ and 50 random initializations. The result was:

Table 2: True classes versus K-means clusters for $K = 3$.

	Cluster 0	Cluster 1	Cluster 2
True Class 1	0	0	20
True Class 2	20	0	0
True Class 3	0	20	0

The cluster numbers are arbitrary, but each true class corresponds exactly to one K-means cluster. The Adjusted Rand Index (ARI) is 1.00, indicating perfect agreement between the cluster structure and the true classes.

(d) K-Means with $K = 2$

With only two clusters, K-means cannot represent the three true groups separately.

Table 3: True classes versus K-means clusters for $K = 2$.

	Cluster 0	Cluster 1
True Class 1	0	20
True Class 2	20	0
True Class 3	20	0

Class 1 remains separate while Classes 2 and 3 are merged. This is an example of under-clustering because the selected number of clusters is smaller than the true number of groups.

(e) K-Means with $K = 4$

With four clusters, K-means is forced to create one more group than actually exists.

Table 4: True classes versus K-means clusters for $K = 4$.

	Cluster 0	Cluster 1	Cluster 2	Cluster 3
True Class 1	20	0	0	0
True Class 2	0	0	20	0
True Class 3	0	8	0	12

Classes 1 and 2 remain intact, while Class 3 is unnecessarily divided into two clusters of 8 and 12 observations. This demonstrates over-partitioning when K is too large.

(f) K-Means with $K = 3$ on only PC1 and PC2

K-means was next applied to the 60×2 matrix containing only the first two principal component scores. It again recovered the three classes perfectly:

Table 5: K-means on the first two PCA scores.			
	Cluster 0	Cluster 1	Cluster 2
True Class 1	0	0	20
True Class 2	20	0	0
True Class 3	0	20	0

The ARI is again 1.00. Thus, in this simulation, PCA reduces the data from 50 variables to only two components without losing the information needed to identify the three classes.

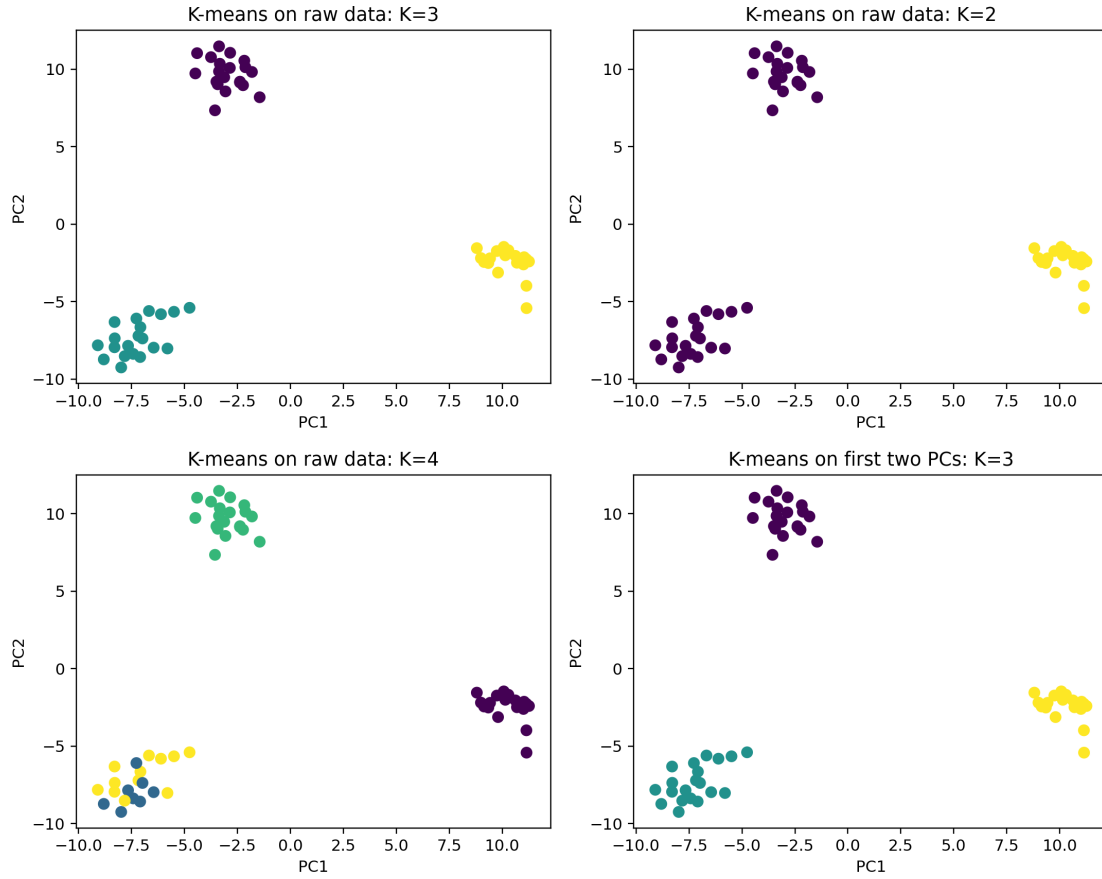


Figure 2: Comparison of K-means results for different choices of K and for PCA-based clustering. All panels are displayed on the same PC1–PC2 coordinates for visualization.

(g) Standardize all variables, then use K-Means with $K = 3$

All 50 variables were standardized to mean zero and standard deviation one using `StandardScaler()`, after which K-means with $K = 3$ was repeated.

Table 6: K-means after standardizing all 50 variables.

	Cluster 0	Cluster 1	Cluster 2
True Class 1	0	0	20
True Class 2	20	0	0
True Class 3	0	20	0

The standardized analysis also gives $\text{ARI} = 1.00$. In this simulated example, the variables already have similar noise variances, so scaling does not alter the main three-group structure. However, Question 5 shows why scaling can matter greatly when variables are measured on very different numerical scales.

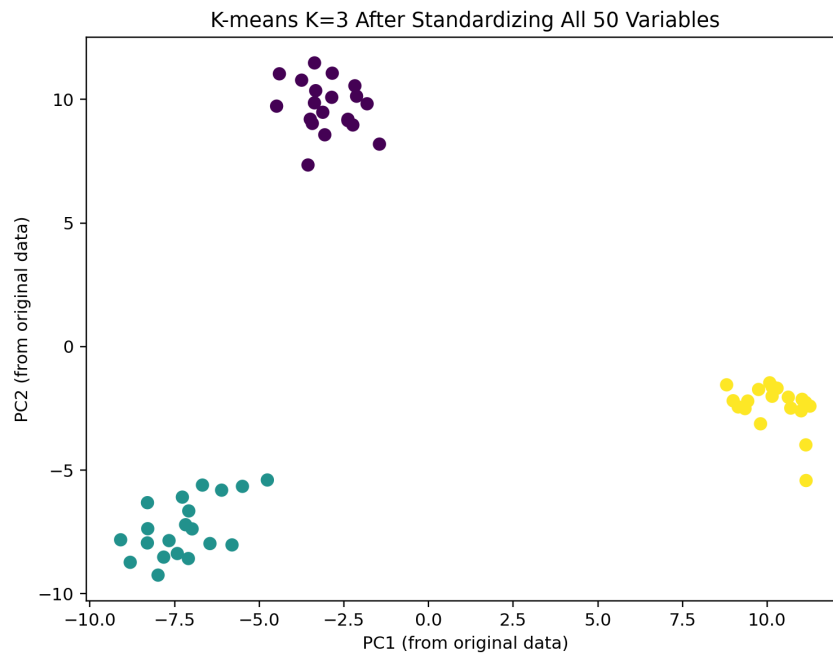


Figure 3: K-means $K = 3$ after standardizing all 50 variables, visualized using PC1 and PC2.

Conclusion

The two exercises illustrate an important principle in unsupervised learning: clustering results depend both on the true structure of the data and on preprocessing choices. In Question 5, changing the scale of the features changes which variable dominates Euclidean distance and therefore changes the clustering. In Question 10, $K = 3$ correctly recovers the three simulated classes, $K = 2$ merges classes, and $K = 4$ splits a true class. PCA retains enough information in two components to preserve the three-group structure in this example.